

Olentangy State Scenic River

2022 Volunteer Monitoring Report

Scenic Rivers Program staff and volunteers monitor the health of Ohio's 15 designated state scenic rivers annually using aquatic macroinvertebrates (or "stream bugs.") In 2022, volunteers conducted biological monitoring across Ohio, collecting a total of 425 samples at 126 locations and contributing 4,436 instream hours. This work is important to ensure that Ohio's State Scenic Rivers continue to remain healthy for future generations to enjoy.

Using Aquatic Macroinvertebrates to Determine Stream Health

The Ohio Scenic Rivers Program developed the Volunteer Stream Quality Monitoring Project in 1983 to encourage communities to monitor their local State Scenic River based on the presence or absence of different types of aquatic macroinvertebrates. Aquatic macroinvertebrates are organisms that lack a backbone (invertebrate), are large enough to be seen with the un-aided eye (macro) and spend a portion of their lives living in water (aquatic). Different types of aquatic macroinvertebrates have varying tolerance levels to aquatic pollution. For example, mayflies are sensitive to pollution, crayfish are somewhat sensitive to pollution and black fly larvae are tolerant of pollution. Results are used to calculate a Cumulative Index Value (CIV) score, which is ranked as excellent, good, fair or poor. Example:

Macroinvertebrate Tally		Letter Codes For Approximate Counts of Each Taxa Type Found: 1-9=A 10-99=B 100+=C				
Sensitive Taxa	Letter Code	Somewhat Sensitive	Letter Code	Tolerant	Letter Code	Cumulative Index Value (CIV)
Water Penny Larvae	A	Damselfly Nymphs	A	Black fly Larvae	B	Stream Quality Assessment Rating
Mayfly Nymph	B	Dragonfly Nymphs		Aquatic Worms	A	
Stonely Nymph		Crane Fly Larvae	A	Midge Larvae	C	
Dobsonfly Larvae		Beetle Larvae	B	Pouch Snails	A	Excellent = CIV > 22
Caddisfly Larvae	C	Crayfish	A	Leeches	A	Good = CIV 17-22
Riffle Beetle Adult	B	Scuds				Fair = CIV 11-16
Gilled Snails		Clams	A			Poor = CIV < 11
		Sowbugs				
Number of Taxa	4	Number of Taxa	5	Number of Taxa	5	Calculated CIV:
4x3	12	5x2	10	5 x1	5	12+10+5= 27 CIV

Summary of the 2022 Monitoring Season

Overall, the Olentangy River scored an average CIV of 26, which is excellent. SQM volunteers took 20 samples and contributed 88 hours of their time to gather this data. From May - October of 2022, the Central Ohio region received above average precipitation and experienced above average temperatures. Volunteers also monitored total suspended solids or turbidity, which is caused by excess soil or organic material and can be harmful to aquatic life. This is particularly noticeable after heavy rainfalls and where the river is not protected by a vegetated border. In 2022, average total suspended solid measurements in the Olentangy were 28 mg/L, indicating normal water quality. Overall, the 2022 season was excellent, and we thank all of those that volunteered their time for this project. We will be looking forward to seeing you again next year!

Thank you SQM Volunteers!

We would like to thank these Ohioans for volunteering their time to monitor the Olentangy River in 2022: John Lorenz, the Lyskey Family, Mark & Brenda Layman, Erin Wolfe, Jim Estep, John Beitel, Will Forister, the Goodalls, Stephanie Vonderwall and Family, Matt Hildebrand, Richer Malboeuf, Dawn Varricchio & Kittie Brown, Rich Skopin, Ed Meixer, and Mike Kohr. The continued success of the Stream Quality Monitoring Project depends on the commitment and dedication of these (and past) volunteers and participants.



Learn more about the Olentangy: <https://ohiodnr.gov/scenicrivers>

Become a Volunteer: contact Christine Szymanski at Christine.Szymanski@dnr.ohio.gov

Olentangy State Scenic River

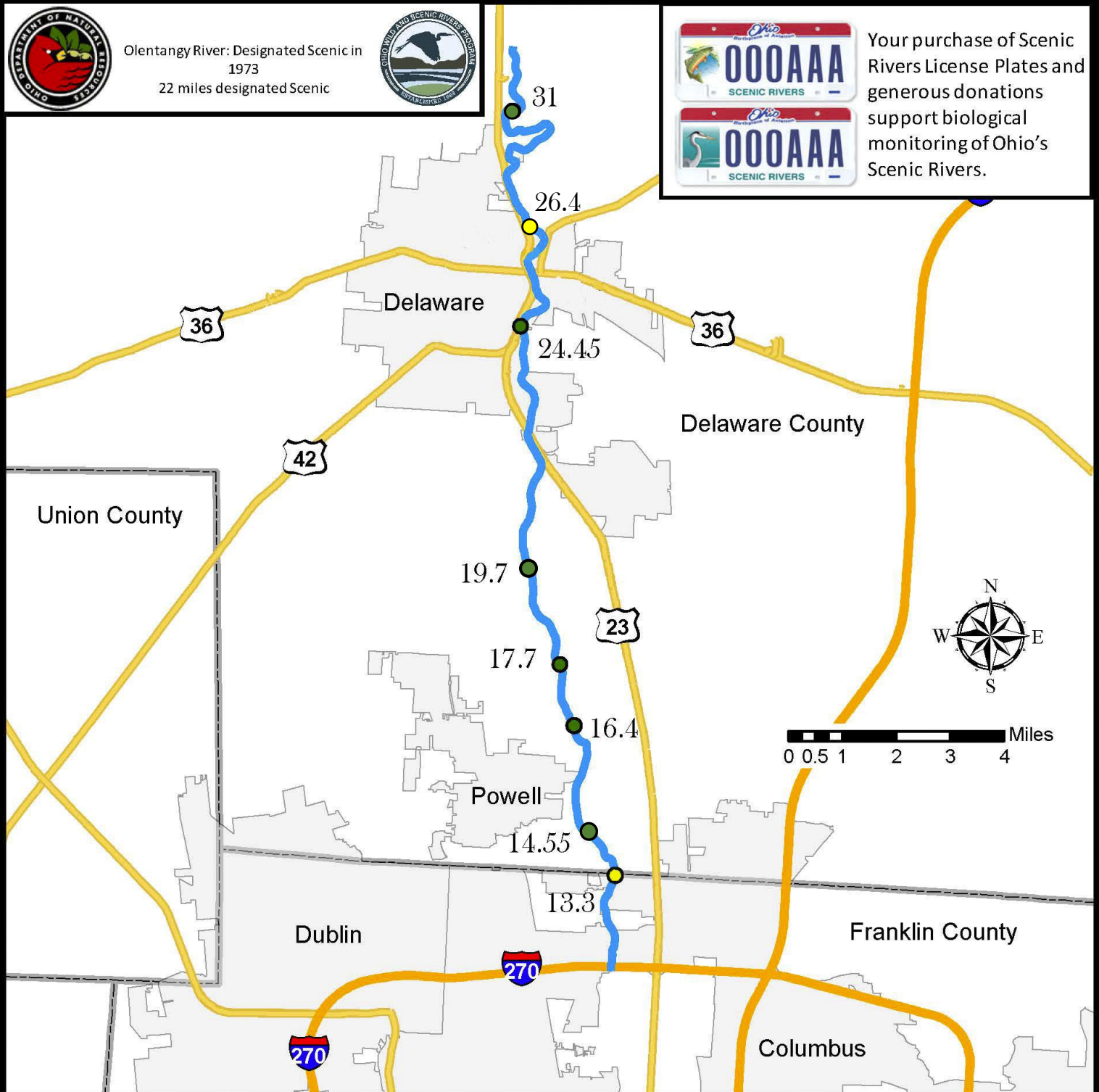
Stream Quality Monitoring Sampling Stations



Olentangy River: Designated Scenic in 1973
22 miles designated Scenic



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Cumulative Index Value (CIV) Score

- 11-16 Fair
- 17-22 Good
- 23-29 Excellent
- 30-42 Excellent

How to read this map: Each colored dot represents a monitoring location. The number next to the dot is the monitored river mile. The color in the dot represents the average score of all the monitoring events on that site within a given year.

Example:



This monitoring location at River Mile 25.8 scored an annual average CIV between 23-29.